# Combinations with Repetition *(bars and stars)*

- The *bars and stars* problem is the same as selecting *r* out of *n* elements of a set, provided that you can select the same element multiple times.
- **Theorem**
  The number of $r$-combinations from a set with $n$ elements when repetition of elements is allowed is
  $C(n+r-1, r) = \frac{(n+r-1)!}{r!(n-1)!}$
- **Proof**: Each such $r$-combination can be represented by a list of $n-1$ bars and $r$ stars. The bars mark the $n$ cells, each corresponding to a distinct element. The stars between two bars correspond to the repetitions of an element.
  The list has $n+r-1$ positions, and we choose $r$ of them to place the stars. Or equivalently, we choose $n-1$ positions to place the bars.